meshthingy Bill of Materials

Source: bom/meshthingy-bom.csv
Rev 95bc552-dirty · 2026-10-10

36 board lines **76** placed parts **4** off-board items **4** open items

☐ tick when kitted · OPEN = decide before ordering

BOARD ASSEMBLY

	#	QTY	REFERENCE	VALUE	DESCRIPTION	MPN / MANUFACTURER	FOOTPRINT	NOTES
☐	1	**10**	C1,C7,C8,C10,C11, C19,C21,C22,C23, C24	**100n**	Unpolarized capacitor	*generic*	C_0805_2012Metric	X7R, >=16 V
☐	2	**5**	C2,C9,C14,C18,C20	**10u**	Unpolarized capacitor	*generic*	C_0805_2012Metric	X5R/X7R, >=10 V
☐	3	**2**	C3,C5	**4u7**	Unpolarized capacitor	*generic*	C_0805_2012Metric	X5R/X7R, >=10 V (C5 on battery, C3 on VSYS)
☐	4	**3**	C4,C6,C17	**1u**	Unpolarized capacitor	*generic*	C_0805_2012Metric	X5R/X7R; C4 is on USB VBUS: use >=16 V
☐	5	**1**	C13	**XH414H-IV01E**	GPS backup cell, rechargeable 3.3 V coin	**XH414H-IV01E** Seiko Instruments	CAP_XH414H-IV01E	GPS backup cell (rechargeable)
☐	6	**2**	C16,C25	**22u**	Tantalum capacitor, 3528	*generic*	CP_EIA-3528-15_AVX-H	OPEN TODO voltage rating: >=10 V low-ESR tantalum (VSYS up to 4.4 V; C25 sees battery hot-plug)
☐	7	**5**	D1,D3,D4,D6,D7	**1N4148WS**	75V 0.15A Fast switching Diode, SOD-323	*generic*	D_SOD-323	D3/D4/D6/D7 power latch: low leakage required, do not substitute Schottky
☐	8	**1**	D2	**LED**	LED, 0805	*generic*	LED_0805_2012Metric	Status LED, any colour (R22 = 1k)
☐	9	**2**	D5,D9	**1N5819WS**	40V 600mV@1A 1A SOD-323 Schottky Barrier Diodes, SOD-323	*generic*	D_SOD-323	Flyback diodes for buzzer (D5) and vibration motor (D9)
☐	10	**1**	J1	**USB_C_16P**	USB 2.0-only 16P Type-C Receptacle connector	**GT-USB-7010ASV** G-Switch	USB_C_Receptacle_G-Switch_GT-USB-7010ASV	
☐	11	**1**	J2	**BATT_Con**	JST-PH 2-pin horizontal, battery	**S2B-PH-K-S** JST	JST_PH_S2B-PH-K_1x02_P2.00mm_Horizontal	
☐	12	**1**	J3	**Conn_Coaxial**	U.FL receptacle, GPS antenna	**U.FL-R-SMT-1** Hirose	U.FL_Hirose_U.FL-R-SMT-1_Vertical	
☐	13	**1**	J4	**SMA_Con**	SMA edge-mount jack, LoRa antenna	**132289** Amphenol RF	SMA_Amphenol_132289_EdgeMount	
☐	14	**1**	L1	**47nH**	Inductor, 0402	*generic*	L_0402_1005Metric	GPS active-antenna bias choke
☐	15	**1**	L2	**6.8uH**	Power inductor, 6x6 mm shielded	*generic*	L_Changjiang_FNR6045S	Buck inductor, Isat >= 1.5 A (e.g. Changjiang FNR6045S6R8MT)
☐	16	**1**	LS1	**Buzzer 8.5x8.5x3**	Magnetic buzzer/transducer, 8.5 x 8.5 mm	**SMT-0927-S-14-R** PUI Audio	XDCR_SMT-0927-S-14-R	OPEN Magnetic transducer, PWM driven - verify exact MPN
☐	17	**2**	Q1,Q4	**SI2302**	N-channel MOSFET, SOT-23 (buzzer / motor driver)	*generic*	SOT-23	N-MOSFET, logic-level (Vgs(th) <= 1.5 V); e.g. Vishay SI2302CDS or equivalent
☐	18	**1**	Q2	**MMBT3906**	PNP transistor, SOT-23 (buck enable switch)	*generic*	SOT-23	Or BC857; SOT-23 pinout B-E-C
☐	19	**1**	Q3	**MMBT3904**	NPN transistor, SOT-23 (MCU power hold)	*generic*	SOT-23	Or BC847; SOT-23 pinout B-E-C
☐	20	**2**	R1,R2	**5k1**	Resistor	*generic*	R_0805_2012Metric	USB-C CC pull-downs (Rd), 1%

	#	QTY	REFERENCE	VALUE	DESCRIPTION	MPN / MANUFACTURER	FOOTPRINT	NOTES
☐	21	**8**	R3,R7,R8,R9,R11, R19,R20,R23	**10k**	Resistor	*generic*	R_0805_2012Metric	R3 = BQ24072 TS bias (no thermistor), must stay 10k
☐	22	**3**	R4,R22,R24	**1k**	Resistor	*generic*	R_0805_2012Metric	R4 = charge current ISET (890 mA), 1% required; R22 = LED; R24 = GPS backup charge
☐	23	**1**	R5	**1k2**	Resistor	*generic*	R_0805_2012Metric	ILIM: USB input current limit ~1.34 A, 1%
☐	24	**2**	R6,R14	**47k**	Resistor	*generic*	R_0805_2012Metric	R6 = charge safety timer (~6.3 h); R14 = power-latch pull-up
☐	25	**4**	R10,R12,R13,R21	**4k7**	Resistor	*generic*	R_0805_2012Metric	I2C pull-ups (R10/R12 main bus, R13/R21 GPS EEPROM bus)
☐	26	**4**	R15,R16,R17,R18	**100k**	Resistor	*generic*	R_0805_2012Metric	R16/R17 = VSYS/2 battery sense divider
☐	27	**1**	U1	**OLED096**	0.96” 128 X 64 MONOCHROME GRAPHIC OLED DISPLAY MODULE - I2C	**DM-OLED096-636** Display Module	MODULE_DM-OLED096-636	0.96" I2C OLED, 4-pin; set JP3/JP4 to match module pinout
☐	28	**1**	U2	**ATGM336H**	GNSS receiver module (GPS/BDS)	**ATGM336H-5N71** Zhongke Microelectronics	ATGM336H-5N	
☐	29	**1**	U3	**TPS62046DGQR**	1.2 A synchronous buck, fixed 3.3 V, MSOP-10 PowerPAD	**TPS62046DGQR** Texas Instruments	DGQ10_1P88X1P57	Thermal pad must be soldered (paste or hand-solder)
☐	30	**1**	U4	**HT-RA62**	LoRa module, SX1262, SPI	**HT-RA62-HF** Heltec Automation	Ai-Thinker-Ra-01-LoRa	OPEN Choose HT-RA62-HF for 863-928 MHz, HT-RA62-LF for 433/470 MHz
☐	31	**1**	U5	**TMHU26**	Lever / jog switch: CW, CCW, push	*generic*	tmhu26	Push also powers the board on (via D7)
☐	32	**1**	U6	**BQ24072RGT**	Li-ion charger with power path, VQFN-16	**BQ24072RGTR** Texas Instruments	VQFN-16-1EP_3x3mm_P0.5mm_ EP1.6x1.6mm	
☐	33	**1**	U7	**ESP32-S3-WROOM-1**	ESP32-S3 module, Wi-Fi + BLE, PCB antenna	**ESP32-S3-WROOM-1-N?R? (TBD)** Espressif	ESP32-S3-WROOM-1	OPEN TODO pick variant (e.g. -N8R2 / -N16R8). Do NOT use 1.8 V "-V" variants: IO47/IO48 are used
☐	34	**1**	U8	**AT24C32A**	I2C EEPROM 32 Kbit, TSSOP-8 (GPS)	**AT24CS32-XHM-T** Microchip	TSSOP-8_4.4x3mm_P0.65mm	32 Kbit I2C EEPROM, TSSOP-8 (symbol is AT24CS32-XHM)
☐	35	**1**	U9	**BME280**	Temperature / humidity / pressure sensor, LGA-8	**BME280** Bosch Sensortec	Bosch_LGA-8_2.5x2.5mm_ P0.65mm_ ClockwisePinNumbering	
☐	36	**1**	U10	**LIS3DSHTR**	3-axis accelerometer, LGA-16	**LIS3DSHTR** STMicroelectronics	LGA16R50P5X3_300X300X100N	

OFF-BOARD ITEMS

	#	QTY	REFERENCE	VALUE	DESCRIPTION	MPN / MANUFACTURER	FOOTPRINT	NOTES
☐	37	**1**	(JP1)	**Vibration motor**	Coin/pager vibration motor, 3 V	*generic*	solder pads	Not a schematic part. Red lead to +3V3 pad, black to Q4 drain pad
☐	38	**1**	(J2)	**Battery**	1S Li-ion/LiPo with JST-PH 2.0 plug	*generic*	-	Not on board. Check plug polarity vs + / - silkscreen before connecting
☐	39	**1**	(J4)	**LoRa antenna**	SMA antenna for chosen band	*generic*	-	Not on board
☐	40	**1**	(J3)	**GPS antenna**	Active GNSS antenna, U.FL	*generic*	-	Not on board. VCC_RF bias is applied via L1